



Building a Fortress of Your Fabric Environment: Security Best Practices for Data Engineers

SQLKonferenz

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SQL KONFERENZ 2026

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Let's connect





**Cybercriminals are
getting more
creative just like us**

Data culture and AI transformation are happening now



78%

of organizations are implementing or developing a company wide data foundation¹

75%

of knowledge workers already using AI at work (doubled in the past 6 months)²

1. The Alation State of Data Culture Report
2. Microsoft 2024 Work Trend Index Annual

DATA SECURITY



Data is most secure if no one can access it.

DATA SECURITY

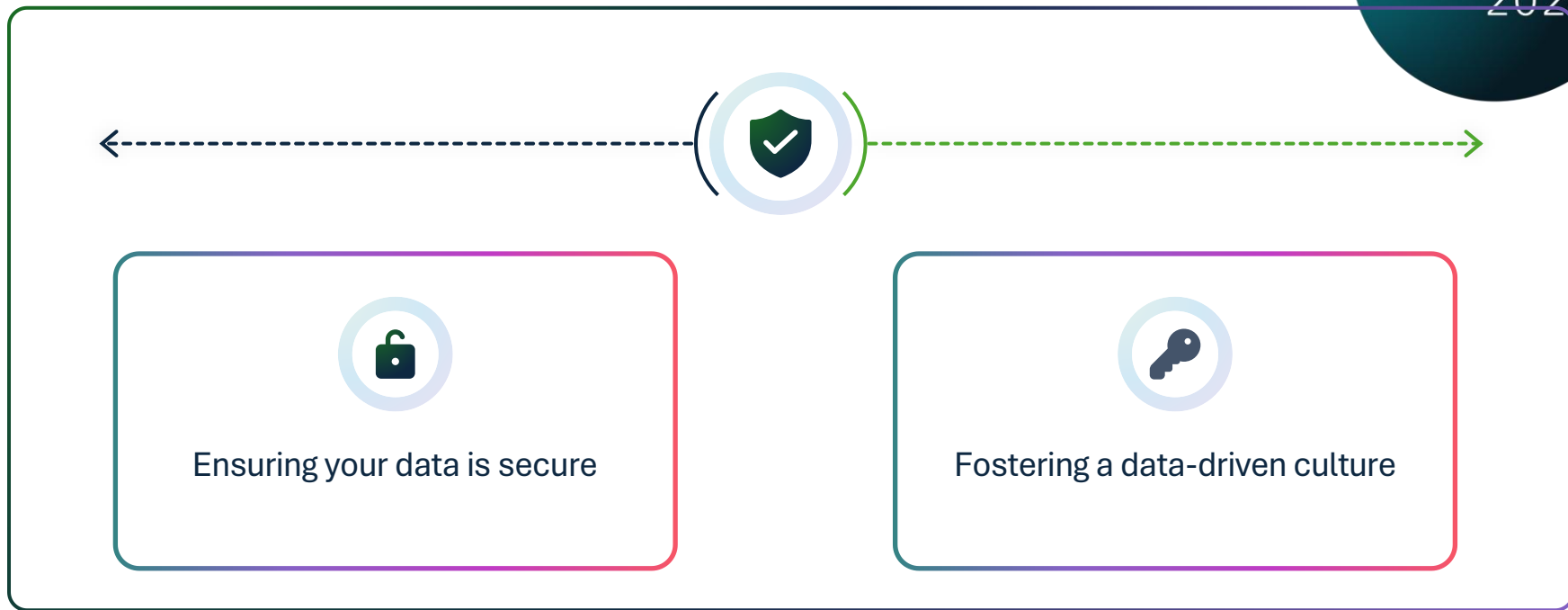


But...if no one can access data then you can't use it for anything.

Achieving both goals can prove challenging



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Agenda

- Introduction
- Challenges
- Inbound Connections
- Private Links
- Outbound connections
- Q & A





- Who has ever heard of the Fabric Security Whitepaper?
- Only 127 pages
- [Microsoft Fabric security white paper - Microsoft Fabric | Microsoft Learn](#)

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AI & Data

A composite image of Earth from space, showing the Americas. A semi-transparent digital grid with glowing dots is overlaid on the lower half of the planet. A horizontal banner with a blue-to-pink gradient is centered across the middle of the image.

AI is transforming the world

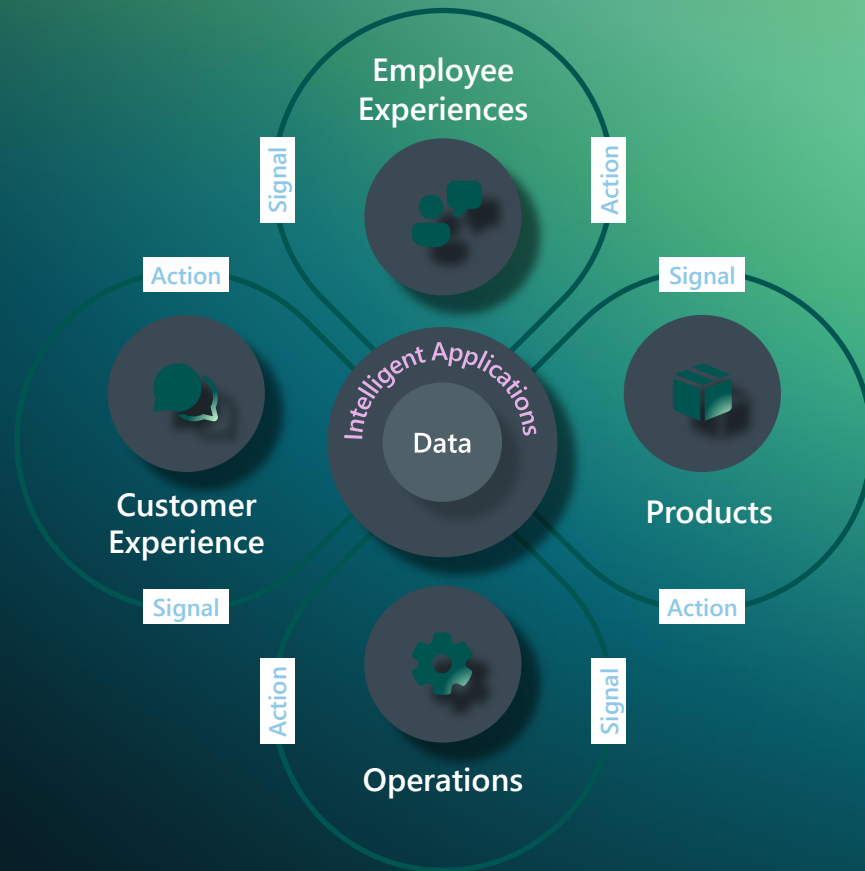
A stylized image of the Earth, showing continents and oceans, with a digital grid overlay. The background is a gradient of green and teal. The text "Data is the fuel that powers AI" is centered over the Earth.

Data is the fuel
that powers AI

Data is the oxygen of our digital transformation

"A new kind of company — we call them insights driven businesses — has formed. They are growing at an average of more than 30% annually"

Forrester Analytics Business Technographics Global Data & Analytics Survey

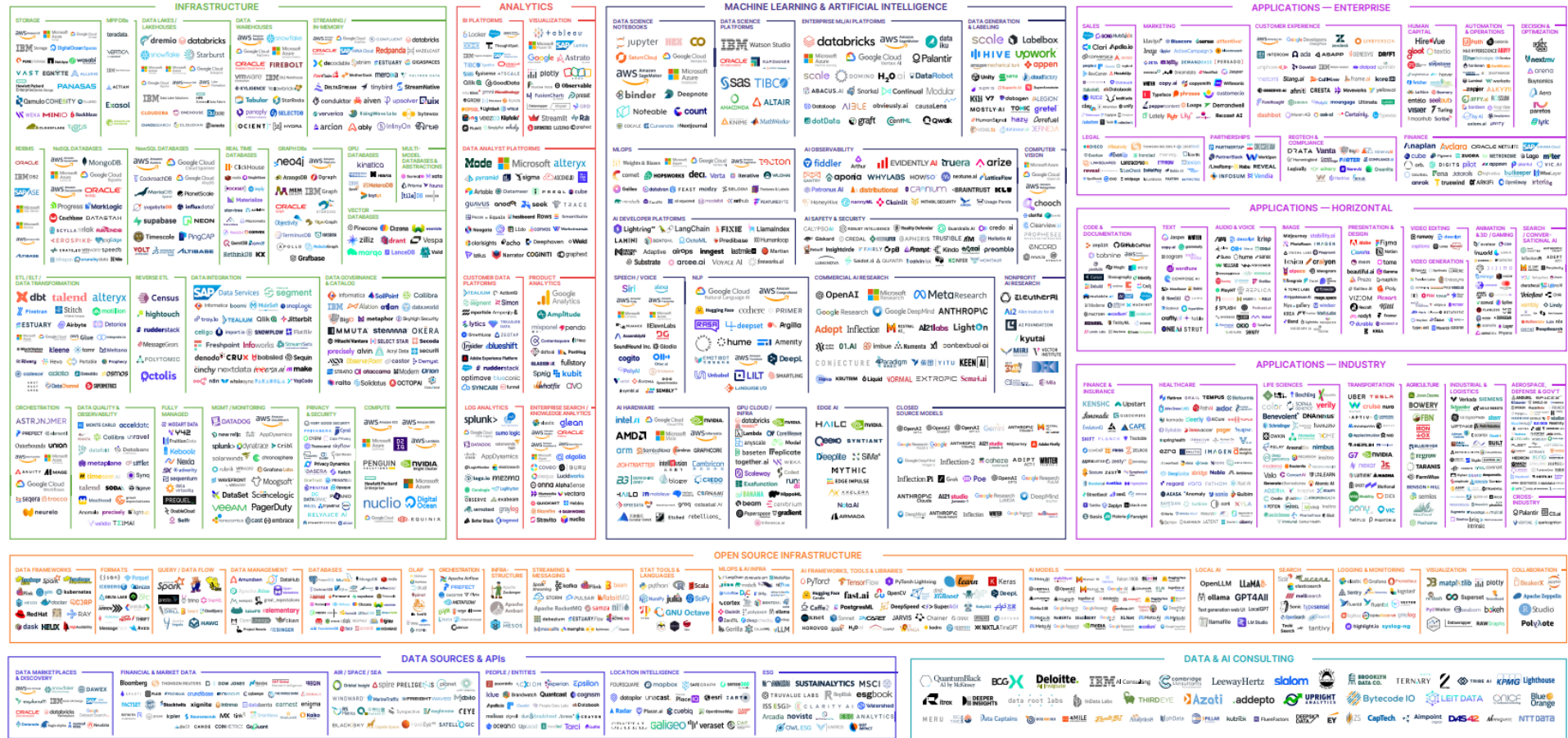


Security : The Cornerstone of Microsoft Fabric

Security is paramount in the digital age, and Microsoft Fabric ensures your data is protected at every stage. By using Fabric, organizations can trust that their data is secure, allowing them to focus on innovation and growth.

The 2025 ML, AI, and Data Landscape

THE 2024 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE





The unified data platform for AI transformation



Data
Factory



Analytics



Databases



Real-Time
Intelligence



IQ



Power BI

Fabric Platform



Copilot



OneLake



Governance

Network security

Workspace & item security

Data security

Your data in Fabric



Regulations &
Certification



Data
Encryption



High Availability
& Disaster
Recovery



E2E
Auditability with
Microsoft
Purview



Information
protection labels
with Microsoft
Purview



Additional
advanced tools
in Microsoft
Purview



3rd party and
in-house
governance
& security
solutions



Network security

Workspace & item security

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Your data in Fabric



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Auditability with
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Information
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with Microsoft
Purview



Additional
advanced tools
in Microsoft
Purview



3rd party and
in-house
governance
& security
solutions

Modern data challenges

- **Bring data to the masses, everyone should have access to data to make decisions**

- **Instant access to the latest data, no copying around**

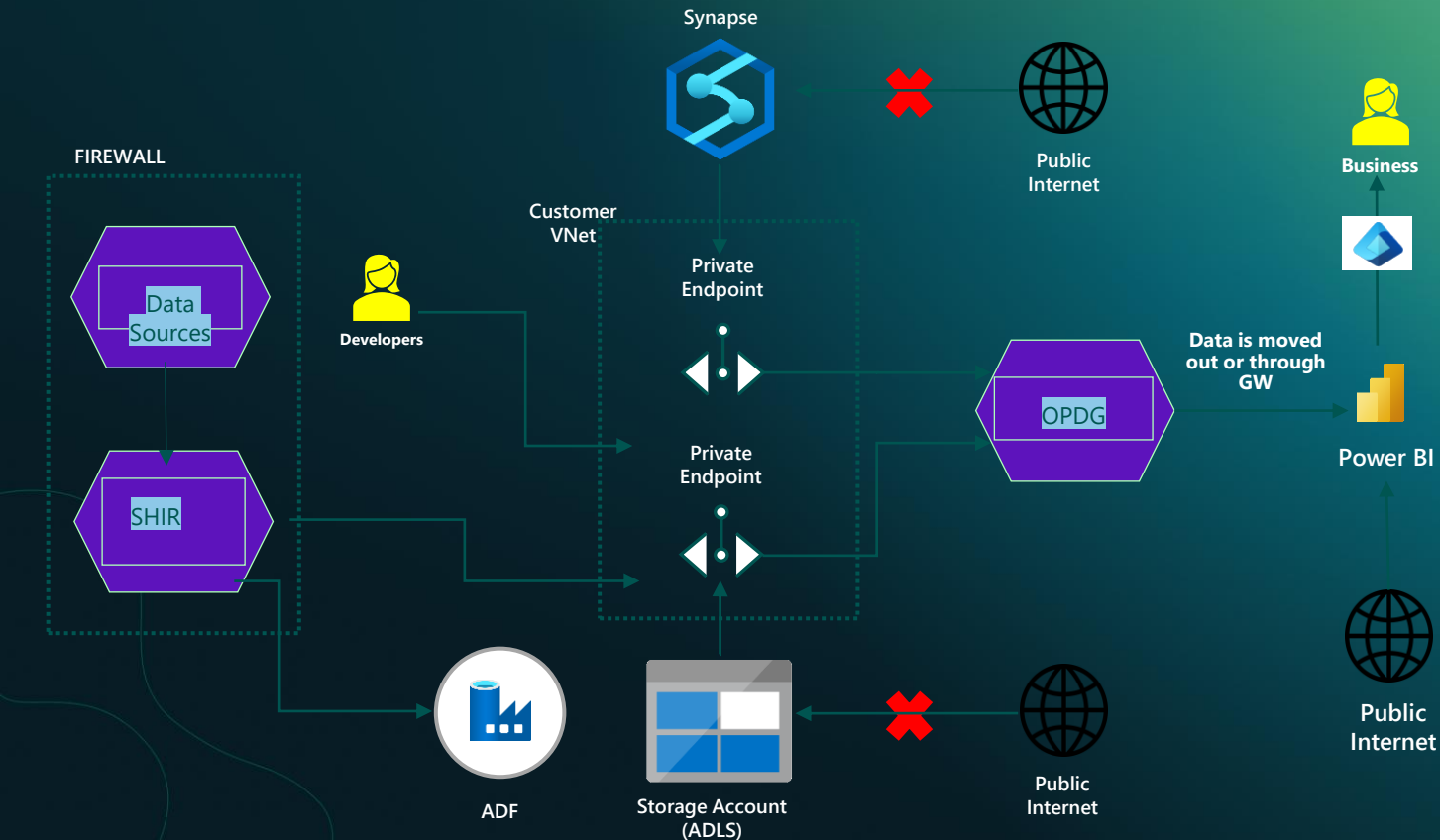
- **Modern workforce, anywhere, any device**

- **At the same time, you need to secure, govern and audit your data to protect customers and the company**

- **SaaS platforms are designed with these challenges in mind.**

- **Shift from siloed PaaS to integrated SaaS**

Existing PaaS World



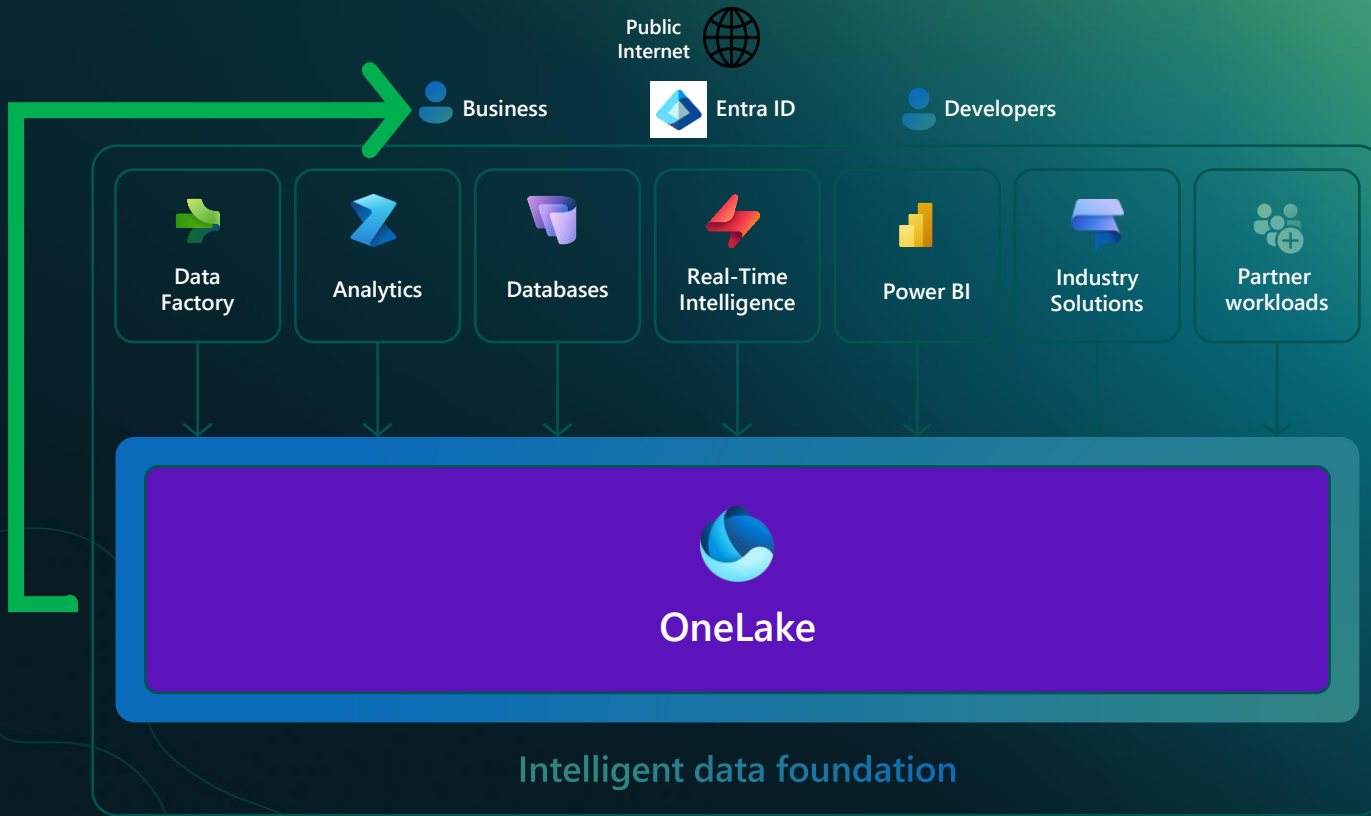
PaaS services are disconnected from the Public internet.

All communications between the services are configured to go through private endpoint

Developers connect to a VPN and connect to the backend services

Data sources are connected through the self-hosted runtime inside the firewall

When the CEO access the data in Power BI Report, the data is imported into Power BI or through the GW with DQ.



Users connect to the SaaS service from global network

Endpoints and access are protected using Entra ID

All internal communications between the experiences happens through MS backbone network

When the CEO access the data in Power BI Report, the data is fetched directly from the OneLake instantly and securely without copying or moving

Common network security requirements



Secure access to your data in Fabric (Inbound Protection)

- Conditional Access Policies
- Private Link at a Tenant Level
- Private Link at a Workspace level
- Workspace IP Firewall



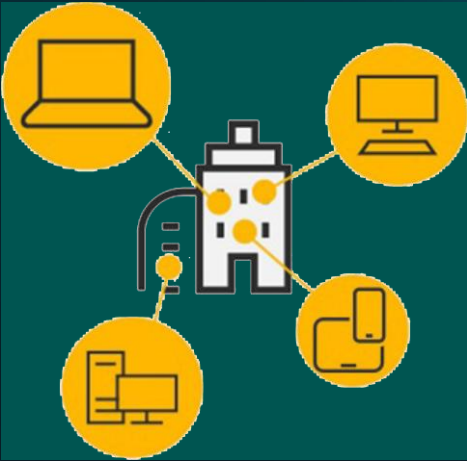
Restrict outbound access from a Fabric workspace (Outbound Protection)



Securely connect to data behind a firewall/private link from Fabric (Outbound Access)

Inbound protection options

Perimeter Network Security



From limited known locations

Zero Trust Approach



to unknown locations

New principles for Zero Trust in today's reality



Verify explicitly



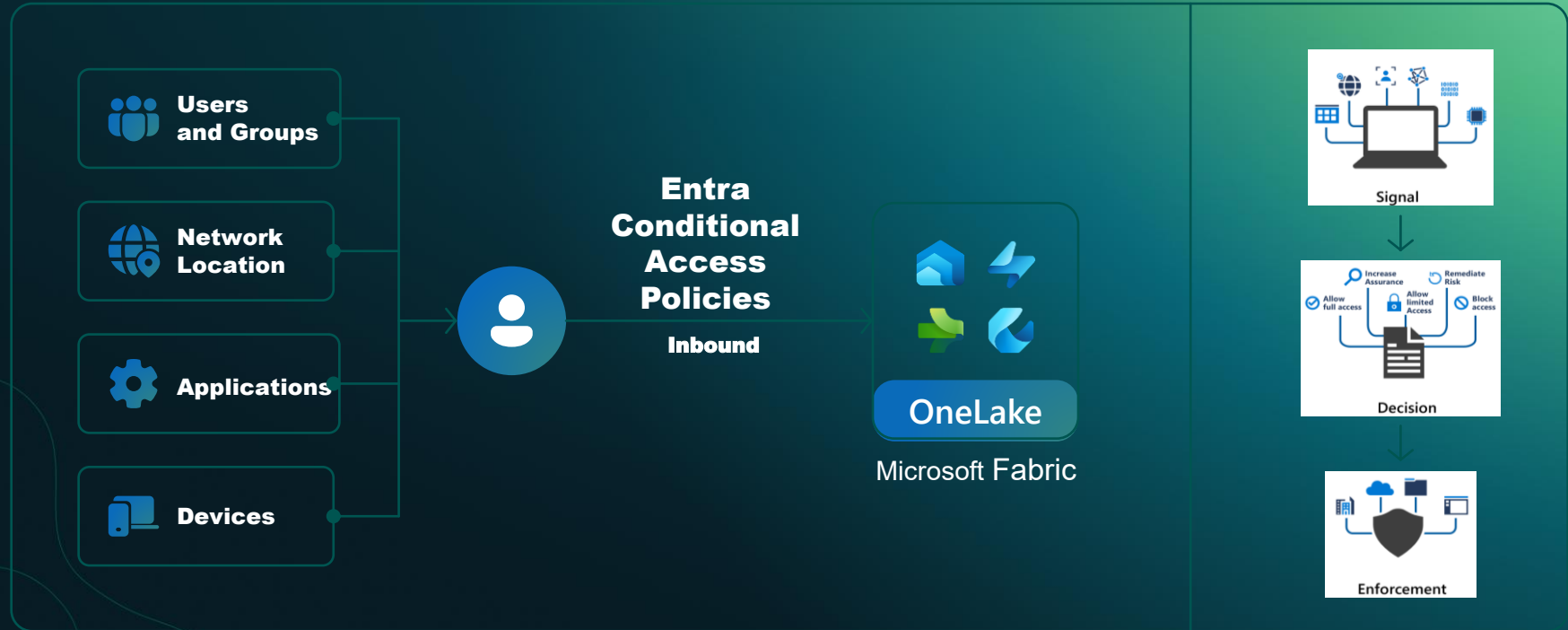
Use least privilege access



Assume breach

Conditional Access Policies

Common Decisions – Block, Grant, Require MFA



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- Applications
- Protection
- Identity Governance
- External Identities
- Show more
- Protection**
- Identity Protection
- Conditional Access
- Authentication methods
- Password reset
- Custom security attributes
- Risky activities
- Show more
- Identity Governance
- Verified ID
- Permissions Management
- Global Secure Access

Microsoft Entra

InSpark Labs
(InSpark B.V.)

Tenant ID cd004ec9-bc4b-4...

Primary domain insparklab...

399

572

239

560

View devices

View apps

Erwin de Kreuk
Global Administrator

a79b59b2-3de7-4c48-b6c8-...

View user profile

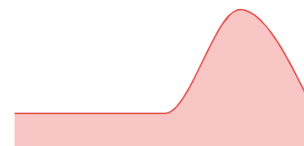
My role assignments



- High privileged role assignments
- Other role assignments

Manage my roles

Users at high risk

8 user detections with risk level
"high" in the last 6 months.[Learn more](#)

75% in the last 30 days

View high risk users

Tenant status

Identity Secure
Score
77.90%

View recommendations

Microsoft Entra
Connect
Enabled

View Entra Connect

Shortcuts

Add

User sign-ins

Audit logs

Authentication Methods

Blocked users

Domain names

Unused service principals

Manage tenants

Named locations

Cross-tenant Access Policies

Tenant restrictions

Risk Based Conditional Access policies

Risky sign-ins

Lifecycle workflows

Get the most out of your licenses and subscriptions

Deployment guides

Use these deployment guides to plan, configure and deploy Microsoft Entra capabilities

Microsoft Entra plan
Entra Suite

- Applications
- Protection
- Identity Governance
- External Identities
- Show more
- Protection
 - Identity Protection
 - Conditional Access
 - Authentication methods
 - Password reset
 - Custom security attributes
- Risky activities
- Show more
- Identity Governance
 - Verified ID
- Permissions Management
- Global Secure Access

Home > Conditional Access

Conditional Access | Named locations

Microsoft Entra ID

Overview

Policies

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

Audit logs

Troubleshooting + Support

New support request

<<

+ Countries location

+ IP ranges location

Configure multiple

Named locations are used by Microsoft Entra security reports to reduce the number of locations marked Trusted or configured in Conditional Access Policies cannot be

All Named Locations

Search by name

3 named locations found

Name	Location type
Block countries	Countries (IP)
The Netherlands	Countries (IP)
Untrusted Countries	Countries (IP)

New location (IP ranges)

Upload Download

Configure named location IPv4 and IPv6 ranges. [Learn more](#)

Name *

☐ Mark as trusted location

Enter a new IPv4 or IPv6 range

ex: 40.77.182.32/27 or 2a01:111::/32

Add

Cancel

Home >



Conditional Access | Overview

Microsoft Entra ID

Overview

[+ Create new policy](#) [+ Create new policy from templates](#)[Refresh](#)[Got feedback?](#)

Policies

Getting started

Overview

Coverage

Monitoring (Preview)

Tutorials

Insights and reporting

Diagnose and solve problems

Manage

[Named locations](#)[Custom controls \(Preview\)](#)[Terms of use](#)[VPN connectivity](#)[Authentication contexts](#)[Authentication strengths](#)[Classic policies](#)

Monitoring

[Sign-in logs](#)[Audit logs](#)

Troubleshooting + Support

[New support request](#)

Policy Summary



Policy Snapshot

23 Enabled 23 Report-only 7 Off

[View all policies](#)

Users

5 users signed in during the last 7 days without any policy coverage

[See all unprotected sign-ins](#)

Devices

100% of sign-ins in the last 7 days were from unmanaged or non-compliant devices

[See all noncompliant devices](#)[See all unmanaged devices](#)

Applications

Browse a list of applications that are not protected by your policies.

[View top unprotected apps](#)

What's new



The approved client app grant in Conditional Access is retiring in March 2026.

This control will no longer be enforced after March 2026. [Update your policy](#) to stay up to date.[Learn more](#)

Named Locations

Microsoft Entra ID now supports IPv6! Update your Named locations with IPv6 ranges.

[Learn more](#)

4 policies have a Named Location condition

- Applications
- Protection
- Identity Governance
- External Identities
- Show more

- Protection**
- Identity Protection
- Conditional Access
- Authentication methods
- Password reset
- Custom security attributes
- Risky activities
- Show more

- Identity Governance
- Verified ID
- Permissions Management
- Global Secure Access

[Home](#) > [Conditional Access | Overview](#) >

New

Conditional Access policy

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies.

[Learn more](#)

Name *

Example: 'Device compliance app policy'

Assignments

Users ⓘ

0 users and groups selected

Target resources ⓘ

No target resources selected

Network **NEW** ⓘ

Not configured

Conditions ⓘ

0 conditions selected

Access controls

Grant ⓘ

0 controls selected

Session ⓘ

0 controls selected

Enable policy

Grant

Control access enforcement to block or grant access. [Learn more](#)

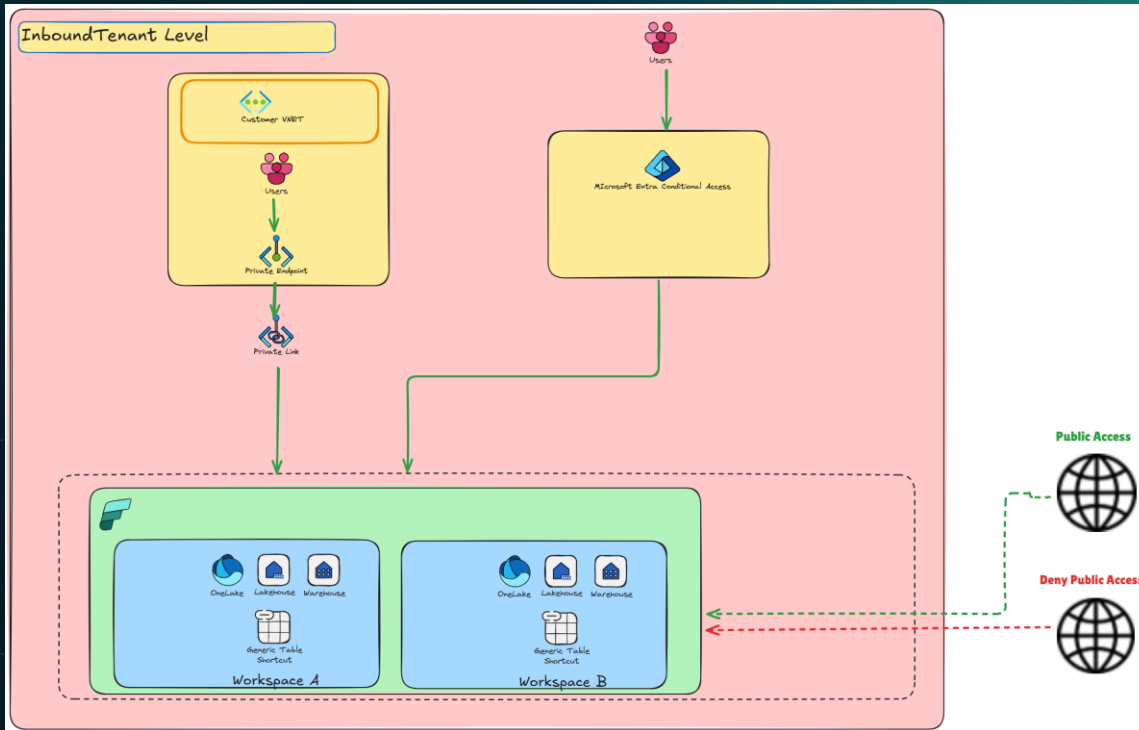
☐ Block access☒ Grant access☐ Require multifactor authentication ⓘ☐ Require authentication strength ⓘ☐ Require device to be marked as compliant ⓘ☐ Require Microsoft Entra hybrid joined device ⓘ☐ Require approved client app ⓘ
[See list of approved client apps](#)☐ Require app protection policy ⓘ
[See list of policy protected client apps](#)☐ Require password change ⓘ☐ All users term of use☐ Guests term of use (Jeffrey TEST)

For multiple controls

☒ Require all the selected controls☐ Require one of the selected controls

Private Link for Fabric Tenant

Perimeter Network Security for your tenant



What it means:

1. Fabric is disconnected from the public internet
2. Every users needs to connect to the private network to get access on every device
3. No longer able to load resources locally (slower reports)
4. Increases ExpressRoute bandwidth and added costs for Private Links
5. Several product limitations (like on-prem data gateway)

Private Link for Fabric Tenant

Perimeter Network Security for your tenant



⏏ Azure **Private Link**

Disabled for the entire organization

Increase security by allowing people to use a **Private Link** to access your Fabric tenant. Someone will need to finish the set-up process in Azure. If that's not you, grant permission to the right person or group by entering their email. [Learn More](#) | [Set-up instructions](#)

Review the [considerations and limitations](#) section before enabling **private** endpoints.

☐ Disabled

🛡 This setting applies to the entire organization

Apply

Cancel

⏏ Block Public Internet Access

Disabled for the entire organization

For extra security, block access to your Fabric tenant via the public internet. This means people who don't have access to the **Private Link** won't be able to get in. Keep in mind, turning this on could take 10 to 20 minutes to take effect. [Learn More](#) [Set-up instructions](#)

☐ Disabled

🛡 This setting applies to the entire organization

Apply

Cancel

Private Link for Fabric Tenant

Perimeter Network Security for your tenant

[Dashboard](#) >

Custom deployment

Deploy from a custom template



Select a template

Basics

Review + create

Automate deploying resources with Azure Resource Manager templates in a single, coordinated operation. Create or select a template below to get started. [Learn more about template deployment](#)



[Build your own template in the editor](#)

Common templates



[Create a Linux virtual machine](#)



[Create a Windows virtual machine](#)



[Create a web app](#)



[Create a SQL database](#)



[Azure landing zone](#)

Start with a quickstart template or template spec

Template source ⓘ



Quickstart template



Template spec

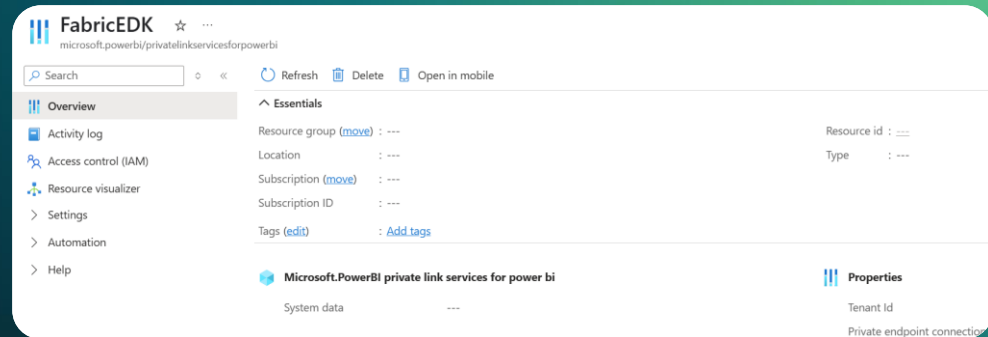
Quickstart template (disclaimer) ⓘ

Private Link for Fabric

Perimeter Network Security for your tenant

- `<resource-name>` is the name you choose for the Fabric resource.
- `<tenant-object-id>` is your Microsoft Entra tenant ID. See [How to find your Microsoft Entra tenant ID](#).

```
{
  "$schema": "http://schema.management.azure.com/schemas/2015-01-01/deploymentschema.json",
  "contentVersion": "1.0.0.0",
  "parameters": {},
  "resources": [
    {
      "type": "Microsoft.PowerBI/privateLinkServicesForPowerBI",
      "apiVersion": "2020-06-01",
      "name": "<resource-name>",
      "location": "global",
      "properties": {
        "tenantId": "<tenant-object-id>"
      }
    }
  ]
}
```



Private Link for Fabric Tenant

Perimeter Network Security for your tenant

Resource instances

Specify resource instances that will have access to your storage account based on their system-assigned managed identity.

Resource type

Instance name

Microsoft.Fabric/workspaces

35bb742f-4dee-437e-b51d-55efaec91cae



Select a resource type



Select one or more instances



Exceptions



Allow Azure services on the trusted services list to access this storage account. ⓘ



Allow read access to storage logging from any network



Allow read access to storage metrics from any network

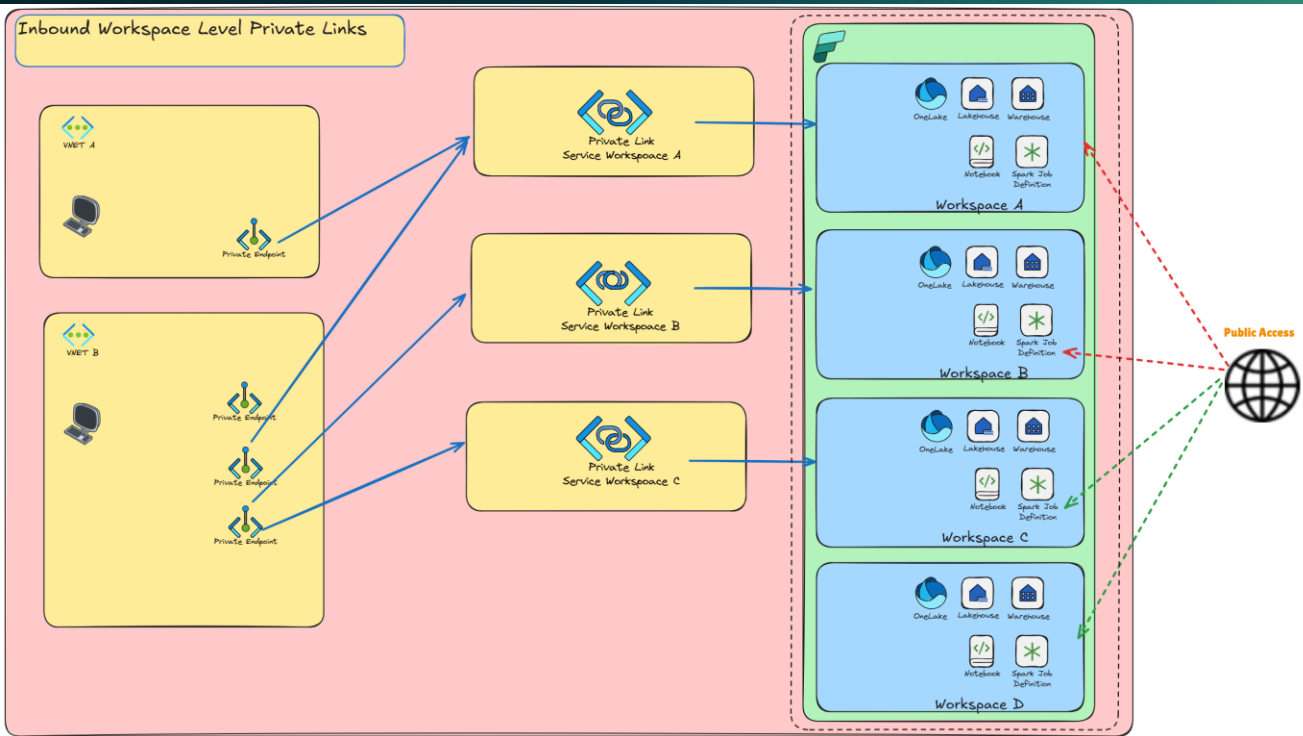
Private Link for Fabric Tenant

Perimeter Network Security for your tenant

- Fabric supports up to 450 capacities, but newly created capacities may take up to 24 hours before private link support becomes active due to private DNS zone propagation.
- Tenant migration is blocked when Private Link is enabled, and you can only connect to one tenant per network location because DNS can resolve only the most recently configured tenant.
- **Trial capacities** aren't supported, Fabric items running on trial capacity will not function when accessed via Private Link.
- On-premises data gateways fail to register when Private Link is enabled; only virtual network data gateways are supported.
- Many features become unavailable, including external images or themes, **Fabric Capacity Metrics app**, OneLake Catalog – Govern tab, and diagnostics for VNet Data Gateway downloads.
- Cross-tenant scenarios aren't supported, each private endpoint can connect to only one tenant, and tenant-level private endpoints cannot connect across Azure tenants.
- Starter Pools are not available when Azure Private Link tenant setting are enabled

Workspace Private Link for Fabric

Perimeter Network Security for your workspace



What it means:

1. Selected workspaces can be protected using Private Links and closed from public internet.
2. Create a secure connection between public and private workspaces using private data access.
3. Public workspaces can be secured using Entra policies or IP filtering to use Power BI.

Workspace Private Link for Fabric

Perimeter Network Security for your workspace



Configure workspace-level inbound network rules (preview)

Unapplied changes

With this setting on, workspace admins can configure inbound **private** link access protection in workspace settings. When a workspace is configured to restrict inbound network access, existing tenant-level **private** links can no longer connect to these workspaces. Turning off this setting reverts all workspaces to their previous configuration. [Learn More](#)

☒ Enabled

 This setting applies to the entire organization

Apply

Cancel

Workspace connection settings

Choose how users and services can connect to this workspace. Use these settings to manage access, security, and integration options. [Learn more](#) 

☒ **Allow all connections to this workspace (including private links)**

Users can connect from any network, including through private links. This setting may be limited by configurations set by your Fabric admin.

☐ **Allow connections only from workspace level private links**

Public access to this workspace is denied when this option is selected.

Apply

Cancel

Workspace Private Link for Fabric

Perimeter Network Security for your workspace

```
{
  "$schema": "http://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {},
  "resources": [
    {
      "type": "Microsoft.Fabric/privateLinkServicesForFabric",
      "apiVersion": "2024-06-01",
      "name": "<resource-name>",
      "location": "global",
      "properties": {
        "tenantId": "<tenant-id>",
        "workspaceId": "<workspace-id>"
      }
    }
  ]
}
```



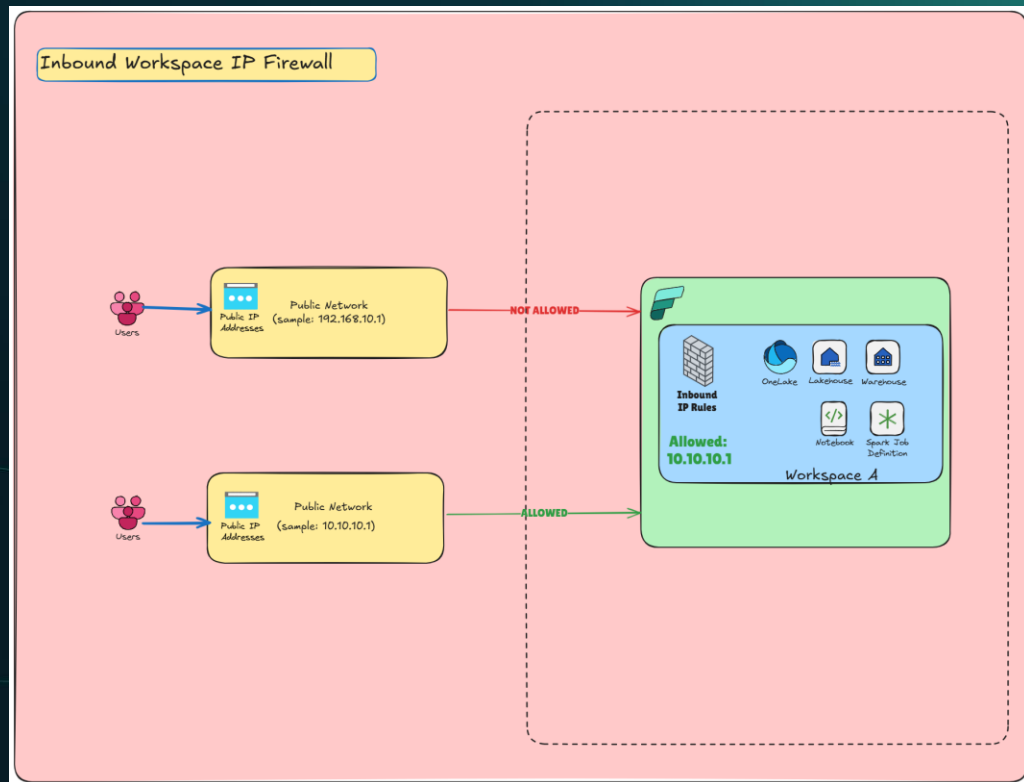
Workspace Private Link for Fabric

Perimeter Network Security for your workspace

- Unsupported item types (such as Deployment Pipelines and Default semantic models) prevent blocking public access; these items cannot be created in a workspace where public access is already restricted.
- Deployment pipelines don't support workspace-level private links, meaning workspaces in a pipeline cannot block inbound public access.
- Inbound restrictions depend on supported items only — if any unsupported item exists in the workspace, public access cannot be fully blocked.
- Admin APIs (admin/workspaces/{workspaceId}) do not support workspace-level private links; only Fabric Core APIs (v1/workspaces/{workspaceId}) operate within the workspace context.

Workspace IP Firewall for Fabric

Perimeter Network Security for your workspace



What it means:

1. Fabric workspace is accessible only from specific public IP or IP Ranges
2. Simple setup.

Fabric Outbound Security



Common network security requirements

→ Secure access to your data in Fabric (Inbound Protection)

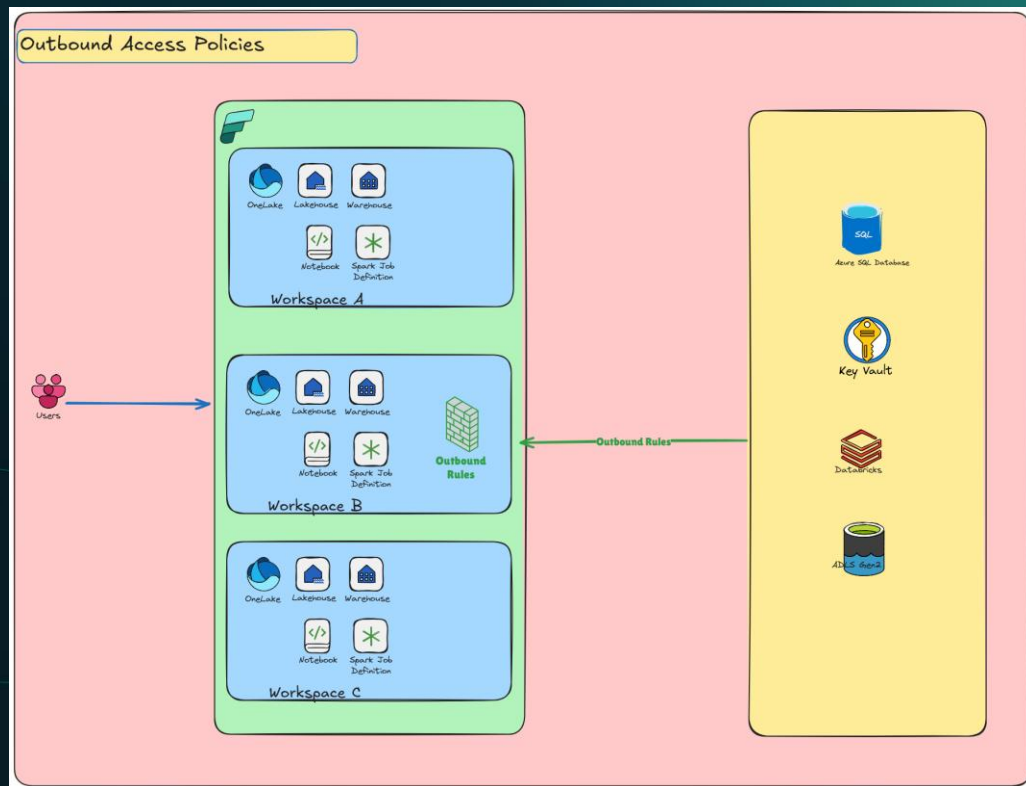
↓ Restrict outbound access from a Fabric workspace (Outbound Protection)

- Outbound Access Policies + Inbound network security features
-

→ Securely connect to data behind a firewall\private link from Fabric (Outbound Access)

Workspace outbound access policies

Outbound Rules



What it means:

1. Restrict outbound connections to permitted destinations only

Workspace outbound access policies

Configure workspace-level outbound **network** rules

Enabled for the entire organization

With this setting on, workspace admins can configure outbound access protection in workspace settings. Turning off this tenant setting also turns off outbound access protection in all the workspaces in the tenant. [Learn More](#)

☒ Enabled

 This setting applies to the entire organization

Apply

Cancel

Data connection rules (preview)

+ Add new cloud connection rule

Save

Cloud connection policies

▼ Azure Data Factory ☒ Allowed 

This connection kind does not support endpoints.

> FabricSqlEndpointMetadata ☒ Allowed 

▼ All other connection kinds ☒ Allowed

This connection kind does not support endpoints.

Gateway connection policies

Workspace outbound access policies

Store data

Organize, query, and store your ingested data in an easily retrievable format.

Cosmos DB database



Build modern apps using fully managed NoSQL database



Datamart (preview)



Provide strategic insights from multiple sources into your business-focused or departmental data.



Eventhouse



Rapidly load structured, unstructured and streaming data for querying.



Lakehouse



Store big data for cleaning, querying, reporting, and sharing.



Sample warehouse



Start a new warehouse with sample data already loaded



Semantic model



Combine data sources in a semantic model to visualize or share it.



Snowflake database (Preview)



Link to a Snowflake database comprised of apache iceberg tables.



SQL database



Build modern cloud apps that scale on an intelligent, fully managed database.



Warehouse



Provide strategic insights from multiple sources into your entire business.



Common network security requirements



Secure access to your data in Fabric (Inbound Protection)



Restrict outbound access from a Fabric workspace (Outbound Protection)

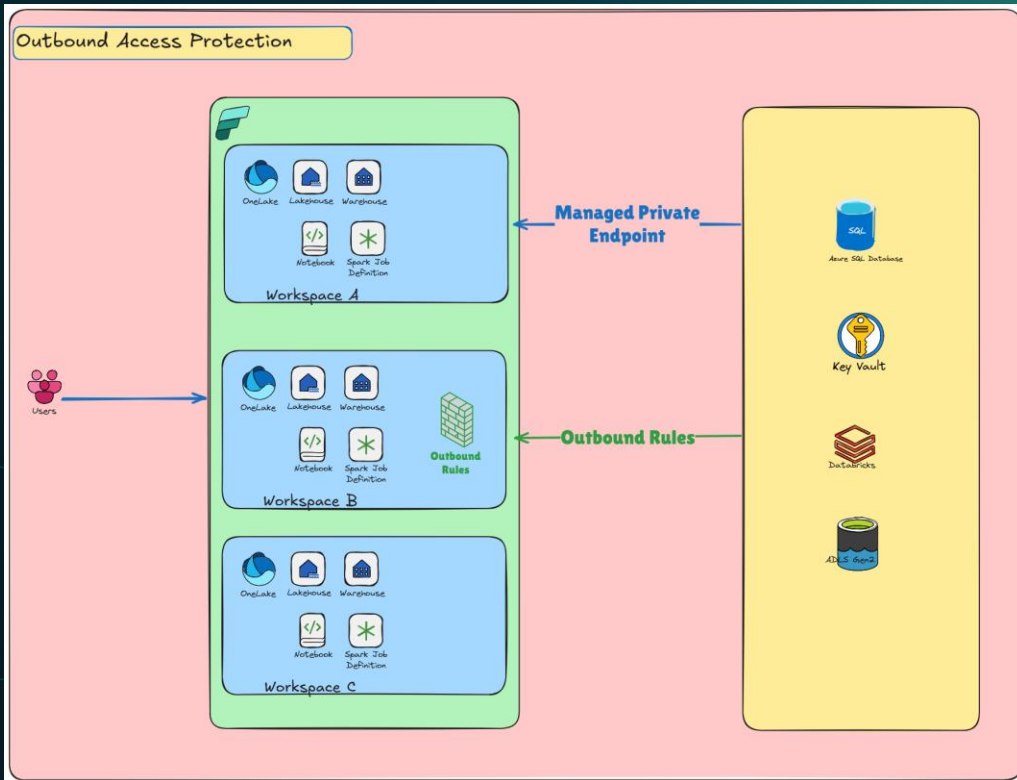


Securely connect to data behind a firewall\private link from Fabric (Outbound Access)

- Trusted Workspace Access
 - Data Gateways(On-prem and VNet)
 - Managed Private Endpoints
-

Getting data into Fabric(Outbound Access Protection)

Managed Private Endpoints



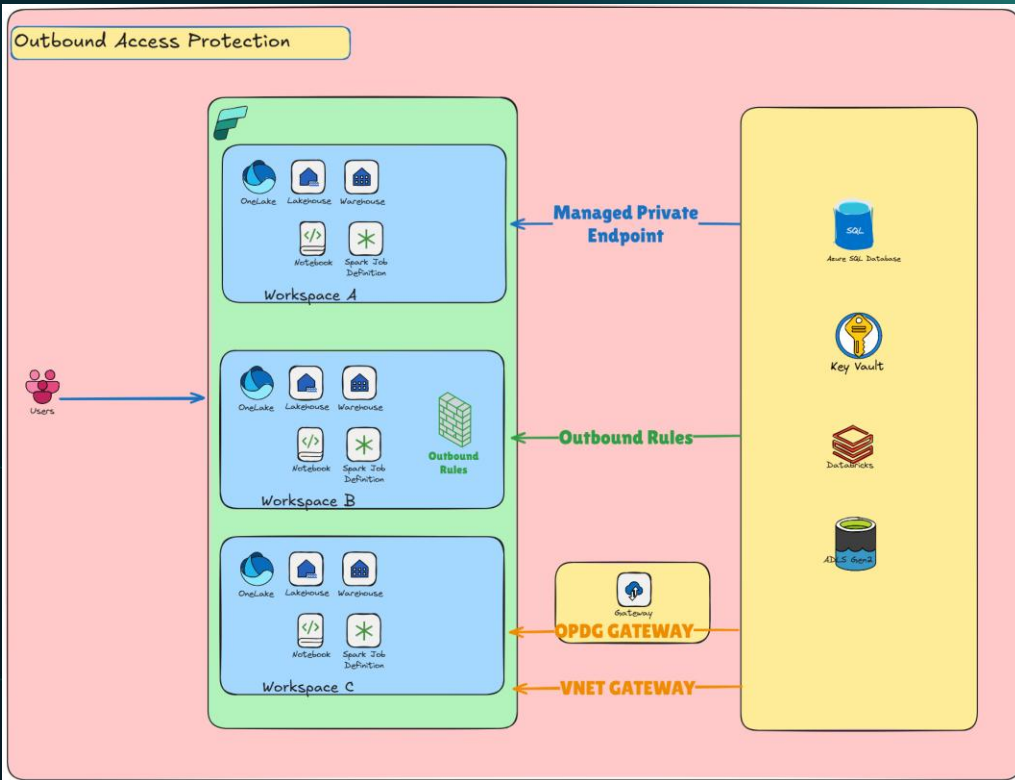
What it means:

1. Create to access data sources that are behind a firewall or that are blocked from public internet access.
2. No complex network configurations
3. Also, possible to On-Prem Datasources

[Connect on-premises data sources to Microsoft Fabric using managed private endpoints](#)

Getting data into Fabric(Outbound Access Protection)

Vnet Gateway or OPDG

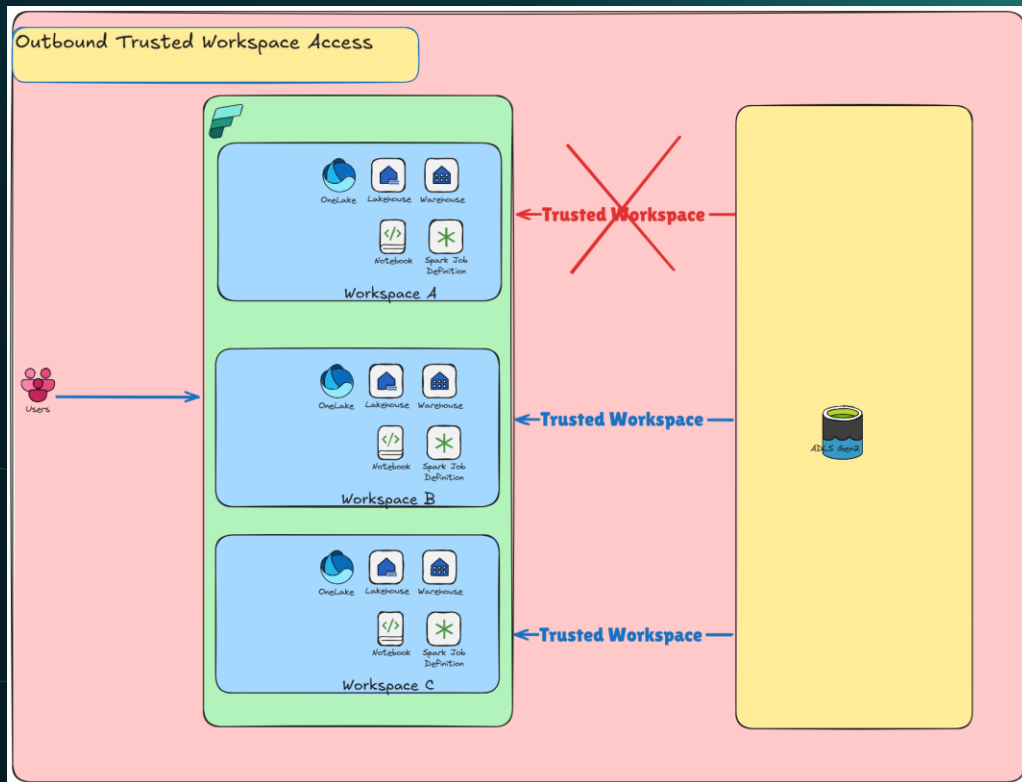


What it means:

1. Create to access data sources that are behind a firewall or that are blocked from public internet access.
2. OPDG or VNET

Getting data into Fabric(Outbound Access Protection)

Trusted Workspace (only for F-SKU)



What it means:

1. Secure access to protected ADLS Gen2 accounts
2. Workspace identity-based authentication
3. Granular workspace-level control

Getting data into Fabric(Outbound Access Protection)

- Storage account name
- Region
- TenantID
- WorkspaceId

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "resources": [
    {
      "type": "Microsoft.Storage/storageAccounts",
      "apiVersion": "2023-01-01",
      "name": "<storage account name>",
      "id": "/subscriptions/<subscription id of storage account>/resourceGroups/<resource group name>/providers/Microsoft.Storage/storageAccounts/<storage account name>",
      "location": "<region>",
      "kind": "StorageV2",
      "properties": {
        "networkAcls": {
          "resourceAccessRules": [
            {
              "tenantId": "<tenantid>",
              "resourceId": "/subscriptions/00000000-0000-0000-0000-000000000000/resourcegroups/Fabric/providers/Microsoft.Fabric/workspaces/<workspace-id>"
            }
          ]
        }
      }
    }
  ]
}
```

Leave subscription for workspace as is

Resource instances

Specify resource instances that will have access to your storage account based on their system-assigned managed identity.

Resource type

Instance name

Microsoft.Fabric/workspaces

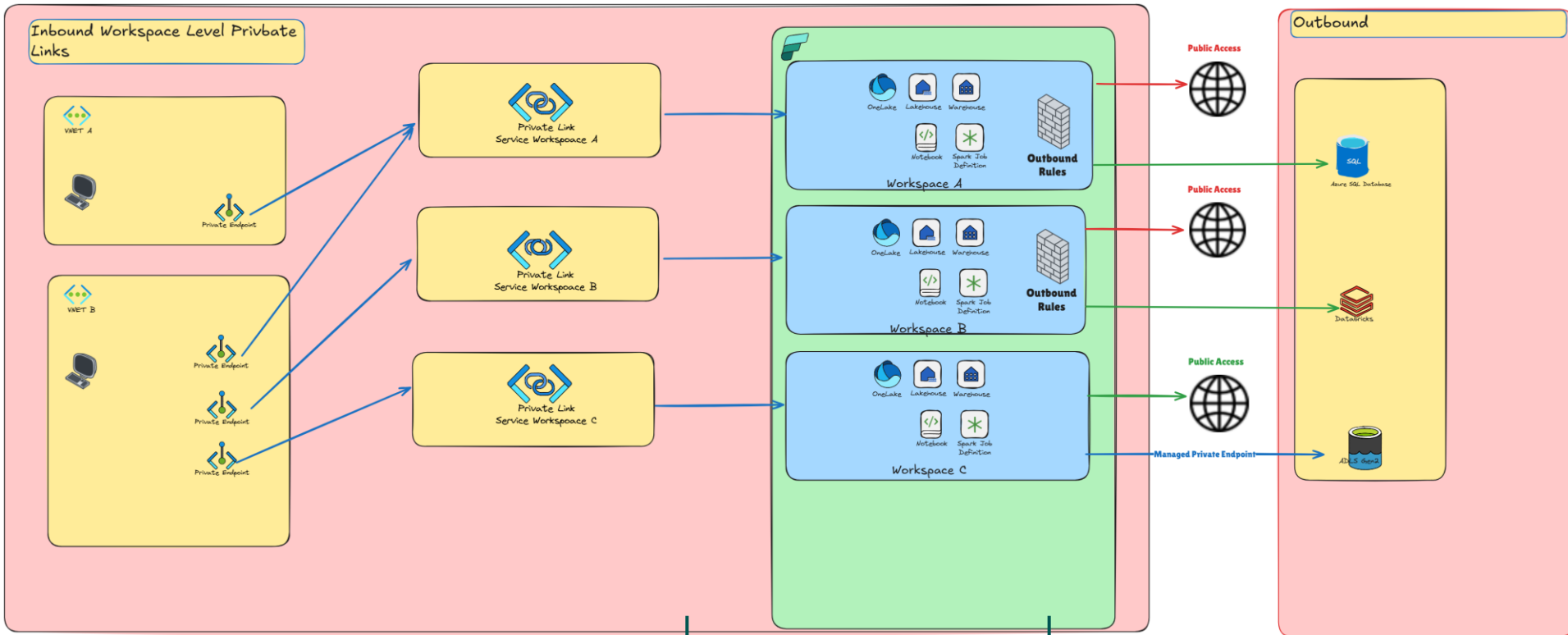
05613f19-4fc0-46d0-add8-72cad691880f

Select a resource type

Select one or more instances

Data Exfiltration Protection

DEP = Outbound + Inbound protection



Inbound security features like private link allow you to restrict who can access your data and from where, reducing risks of data exfiltration



Outbound security make sure that malicious users cannot exfiltrate data

RECAP



Secure access to your data in Fabric (Inbound Protection)

- Conditional Access Policies
 - Private Link at a Tenant Level
 - Private Link at a Workspace level
 - Workspace IP Firewall
-



Restrict outbound access from a Fabric workspace (Outbound Protection)

- Outbound Access Policies + Inbound network security features
-



Securely connect to data behind a firewall\private link from Fabric (Outbound Access)

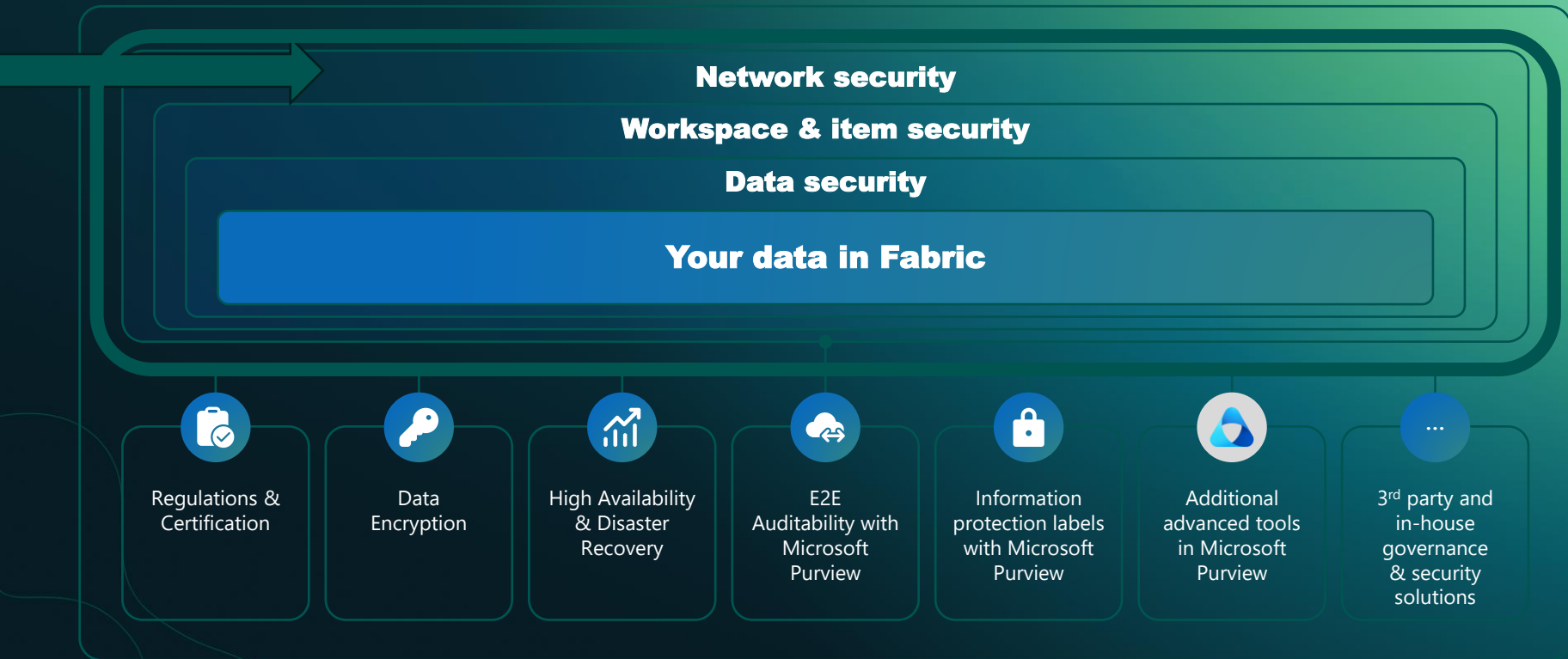
- Trusted Access
 - Data Gateways(On-prem and VNet)
 - Managed Private Endpoints
-

Flexibility with no compromise to Security



DEMO

Recap



Erwin de Kreuk

Technology Lead Data
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FEEDBACK

